

Lab02 tasks

Task1

Q1. A DHCP server keeps a record of leased IP addresses. Before assigning a new address, it selects an unused IP from its configured address pool, so the same IP is not normally given to two devices at the same time.

Q2. For 50 devices, I would suggest a private IPv4 network such as 192.168.1.0/26. A /26 subnet provides 62 usable host addresses, which is enough for 50 devices while avoiding unnecessary address waste.

Q3. I would check the following:

1. Check cables and link lights.
2. Verify both devices have correct IP addresses.
3. Check that both devices use the correct subnet mask.
4. Make sure they are in the same subnet if no router is involved.
5. Use ping 127.0.0.1 to test TCP/IP locally.
6. Ping the device's own IP address.
7. Ping the other device.
8. Check switch ports and VLAN configuration.
9. Check firewall settings if necessary.

Q4. The conflict can be resolved by excluding the statically assigned IP address from the DHCP pool. Another option is to change the static IP to an address outside the DHCP range. This prevents DHCP from assigning that address to another device.

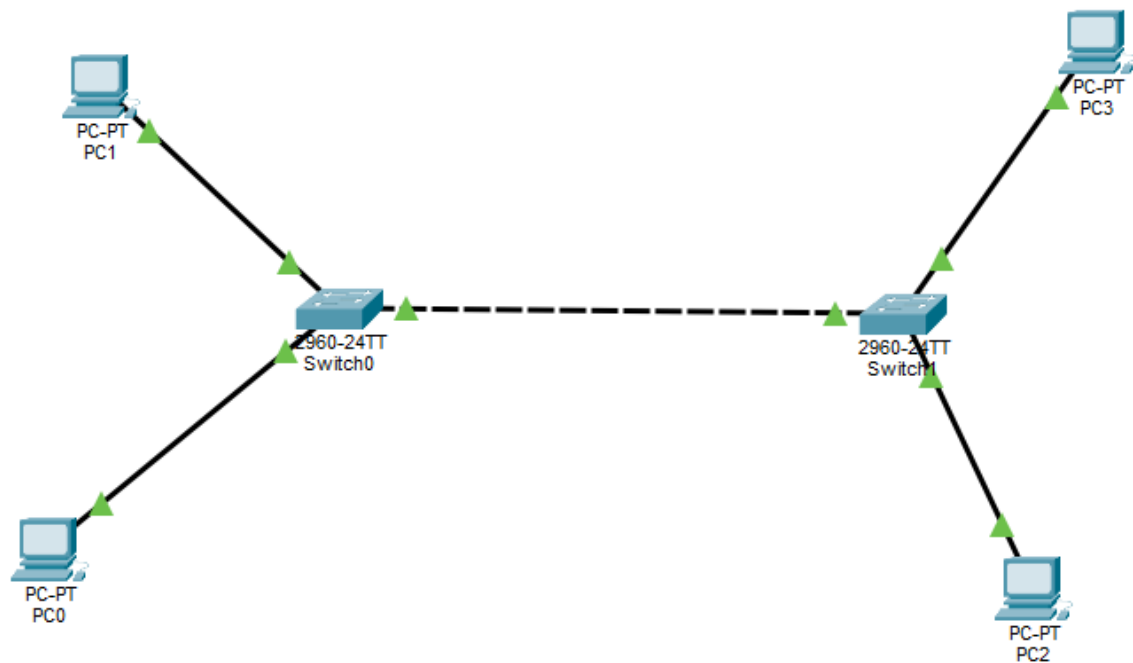
Q5. A switch would provide much better performance than a hub. A hub sends every frame to all connected computers, so all devices share the same bandwidth and collision domain. A switch sends frames only to the correct destination port, reducing collisions and giving each device better bandwidth.

Q6. When a computer connects for the first time, DHCP generally follows the **DORA** process:

1. **Discover** – The computer broadcasts a DHCP Discover message looking for a DHCP server.
2. **Offer** – The DHCP server offers an available IP address and other network settings.
3. **Request** – The computer requests the offered IP address.
4. **Acknowledge** – The DHCP server confirms the lease and sends the final settings.

After this, the computer can use the assigned IP address to communicate on the network.

Task2



Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical

Physical

x: 938, y: 672

Root

03:54:30

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.2

Pinging 192.168.0.2 with 32 bytes of data:

Reply from 192.168.0.2: bytes=32 time<1ms TTL=128
Reply from 192.168.0.2: bytes=32 time<1ms TTL=128
Reply from 192.168.0.2: bytes=32 time=3ms TTL=128
Reply from 192.168.0.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 0ms

C:\>|
```

☐ Top

Time: 00:07:51

Realtime

Simulation

Search for device

Copper Cross-Over

Scenario 0

New Delete

gle PDU List Wind

Fire

Last Status

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 914, y: 753 Root 05:57:30

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 192.162.0.101

Pinging 192.162.0.101 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.162.0.101:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.0.101

Pinging 192.168.0.101 with 32 bytes of data:

Reply from 192.168.0.101: bytes=32 time<1ms TTL=128
Reply from 192.168.0.101: bytes=32 time<1ms TTL=128
Reply from 192.168.0.101: bytes=32 time<1ms TTL=128
Reply from 192.168.0.101: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.0.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.0.102

Pinging 192.168.0.102 with 32 bytes of data:

Reply from 192.168.0.102: bytes=32 time<1ms TTL=128
Reply from 192.168.0.102: bytes=32 time<1ms TTL=128
Reply from 192.168.0.102: bytes=32 time<1ms TTL=128
Reply from 192.168.0.102: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.0.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Time: 00:11:57

Realtime Simulation

Scenario 0

New Delete

gle PDU List Wind

Copper Cross-Over

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 795, y: 158 Root 07:19:00

PC2

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 192.168.01.1

Pinging 192.168.01.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:

Reply from 192.168.0.1: bytes=32 time<1ms TTL=128
Reply from 192.168.0.1: bytes=32 time=3ms TTL=128
Reply from 192.168.0.1: bytes=32 time=3ms TTL=128
Reply from 192.168.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\>ping 192.168.0.2

Pinging 192.168.0.2 with 32 bytes of data:

Reply from 192.168.0.2: bytes=32 time<1ms TTL=128
Reply from 192.168.0.2: bytes=32 time<1ms TTL=128
Reply from 192.168.0.2: bytes=32 time<1ms TTL=128
Reply from 192.168.0.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

PC-PT PC3

PC-PT PC2

Time: 00:14:39

Realtime Simulation

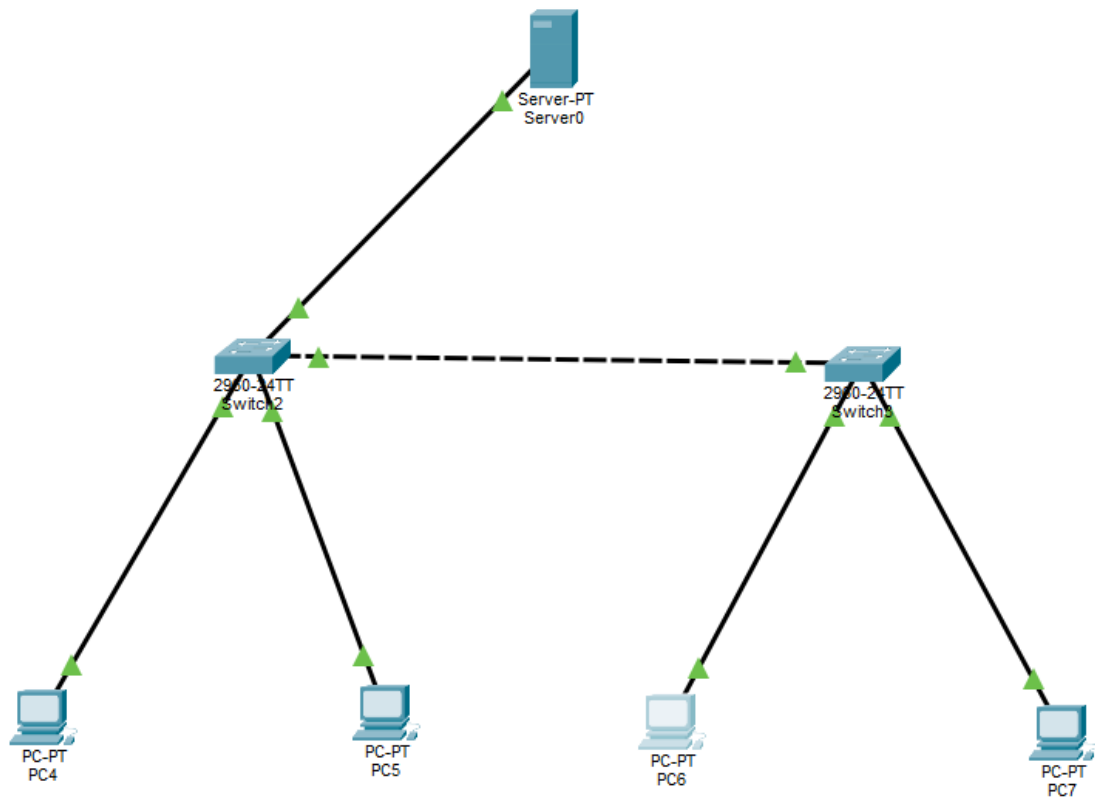
Scenario 0

New Delete

gle PDU List Wind

Copper Cross-Over

Task3



Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 714, y: 1338 Root 10:36:00

PC4

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=13ms TTL=128
Reply from 192.168.1.4: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

PC-PT PC4

Time: 00:21:10

Realtime Simulation

Scenario 0

New Delete

gle PDU List Wind

Copper Straight-Through

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 844, y: 1345 Root 10:49:30

PC4

Physical Config Desktop Programming Attributes

Command Prompt

```
Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=13ms TTL=128
Reply from 192.168.1.4: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 4ms

C:\>ping 192.168.1.100

Pinging 192.168.1.100 with 32 bytes of data:
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

PC-PT PC4

Time: 00:21:36

Realtime Simulation

Scenario 0

New Delete

gle PDU List Wind

Copper Straight-Through

